

■ The ML Project Playbook — Living Document

My personal, growing reference for machine-learning projects.

Every time I learn a new technique or make a decision, I add it here.

Next project, I check this FIRST instead of re-deriving everything.

How to use: before each phase of a new project, open the matching section.

After each project, add anything new to the **Techniques Log** and **Scoreboard** at the bottom.

Last updated: 2026-06-27 · Projects logged: India Cancer, P1 Heart Disease, Telco Churn

Planned queue: ~~Telco Churn~~ ■ → Ames → Fetal Health (CTG) → Insurance → Covertypes → Energy Efficiency → Glioma Grading

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1. Data Loading & Quality

Step	What to do
Shape & dtypes	<code>df.shape</code> , <code>df.dtypes</code> — know rows × cols and which are numeric vs object
Peek	<code>df.head()</code> , <code>df.sample(5)</code>
Missing values	<code>df.isnull().sum()</code> — note which columns, how many
Duplicates	<code>df.duplicated().sum()</code> — exclude ID columns before counting (an ID makes every row unique)
Target balance	<code>y.value_counts(normalize=True)</code> — decides metric (balanced → accuracy ok; imbalanced → F1/PR)
Data realism	Are values plausible? Suspiciously uniform counts → likely synthetic data

2. Data Cleaning

Missing values — decision order

- **Understand WHY it's missing** (value counts, is it related to another column?)

- **Don't drop rows** unless missingness is tiny AND random.

- **Choose imputation:**

Strategy	Use when
median	Numerical, robust to outliers (default for numeric)
most_frequent	Categorical
Grouped imputation	When another column predicts the missing one → fill with that group's median/mode
PowerTransformer/model-based	Advanced, rarely needed early

■ **Grouped imputation (learned in P1 & Titanic):** fill a missing column using the value typical for a related group.

- Titanic: missing Age → median age of each Title group (Master = young boys).
- P1 Heart: missing ca/thal → median/mode of each **age band** (blocked vessels rise with age).
- **Rule:** only helps if the grouping variable actually relates to the missing column — **check the evidence first** (group-wise summary vs global).
- **Build it as a custom transformer inside the Pipeline** so it re-fits per CV fold → no leakage.

Outliers

- Detect with IQR boxplots ($Q1 - 1.5 \cdot IQR$, $Q3 + 1.5 \cdot IQR$).
- **KEEP** if clinically/physically real (e.g. cholesterol 564 is real). Only remove true data-entry errors.
- If outliers worry you, use RobustScaler (median-based) instead of StandardScaler.

Data leakage columns (drop these!)

- Any column that wouldn't exist at prediction time.
- India Cancer: dropped Survival_Months (encodes time-to-death → fake 97% accuracy).
- **Rule:** if a feature is a **consequence** of the target, it's leakage.

3. EDA

Univariate (one variable at a time)

- Histograms — see distribution shape.
- **Skewness** (`.skew()`): $|\text{skew}| > 1$ = highly skewed. **Kurtosis** = tailedness.
- Boxplots — spot outliers.

Bivariate (feature vs target)

- Numerical vs target → boxplots split by class + **t-test**.
- Categorical vs target → bar chart of target rate per category + **chi-square**.
- Numerical vs numerical → scatter + **correlation**.
- **Best practice:** put the plot AND the statistical test result on the same figure (pattern + "is it real?").

Correlation

- **Pearson** = linear relationship. **Spearman** = monotonic (robust to non-linearity).
- Heatmap to spot feature-feature redundancy (multicollinearity).
- ■■ **Sign vs strength:** a correlation of -0.43 is **STRONG** (negative = direction, not weakness). Only $|p|=0$ means weak.

4. Statistical Analysis

Which test for which data type

Feature	Target	Test	Measures
Categorical	Categorical	Chi-square	Are category distributions different?

Feature	Target	Test	Measures
Numerical	Binary (2 groups)	t-test	Are the two means different?
Numerical	3+ groups	ANOVA	Are means different across groups?
Numerical	Numerical	Pearson / Spearman	Do they move together?
Numerical (non-normal)	Categorical	Mann-Whitney U	Non-parametric t-test

Always pair significance with effect size

- p-value answers "is it real?" — **effect size** answers "is it big enough to matter?"
- With large N, tiny meaningless differences become "significant." Effect size guards against this.
- Effect sizes: **Cohen's d** (t-test), η^2 (**eta-squared**) (ANOVA), **Cramér's V** (chi-square), **|p|** (correlation).

Multicollinearity — VIF

- **VIF (Variance Inflation Factor)**: how much a feature is explained by the *other* features.
- VIF < 5 = fine; 5–10 = moderate; >10 = serious redundancy → consider dropping one.

5. Feature Engineering

Technique	Example
Extract from dates	Diagnosis_Date → year, month
Binning	Age → age groups (captures non-linear effects)
Ordinal scores	Treatment type → aggressiveness score 1–5 (domain knowledge)
Interaction features	Stage × Cancer_Type (Stage IV lung ≠ Stage IV breast)
Ratio features	rooms / households
Binary flags	high-risk = (Stage IV AND Palliative) — only if the group is class-pure , check scatter first

Rule: features encode domain knowledge; algorithms find patterns. Better features usually beat more tuning.

Zero-inflated columns (big spike at 0): log won't help. A binary "is-zero" flag only helps if the zero group is mostly one class — verify on the scatter (P1 o1dpeak had both classes at 0 → no flag).

Skewed features — decision

- **Model first:** trees are skew-immune (don't transform); linear models benefit.
- Smooth right tail, all positive → log1p. Has zeros/negatives → PowerTransformer (Yeo-Johnson).
- **Measure, don't assume:** compare raw vs transformed by CV. If within ~0.005, keep it simple.

6. Preprocessing & Pipelines

The leakage rule (most important)

Split FIRST. Fit preprocessing on TRAIN only. Transform both.

Anything that *learns a statistic* (imputation median, scaling mean/std, encoder categories) must be fit inside the Pipeline so CV re-fits it per fold.

Safe before split (row-by-row, learns nothing): dropping columns, extracting date parts, ratios.

The reusable skeleton

```
num_pipe = Pipeline([('imputer', SimpleImputer(strategy='median')),
                    ('scaler', StandardScaler())])
cat_pipe = Pipeline([('imputer', SimpleImputer(strategy='most_frequent')),
                    ('ohe', OneHotEncoder(handle_unknown='ignore'))])
preprocessor = ColumnTransformer([('num', num_pipe, num_cols),
                                  ('cat', cat_pipe, cat_cols)])
full = Pipeline([('pre', preprocessor), ('clf', LogisticRegression())])
```

Encoding

Method	Use when
OneHotEncoder	Low-cardinality categoricals (correct for CV; use <code>handle_unknown='ignore'</code>)
<code>pd.get_dummies</code>	Quick EDA only — NOT inside CV (no train/test consistency)
Ordinal encoding	Ordered categories (quality: low<med<high)
Target / frequency encoding	High-cardinality (many categories) — *to learn in Telco/Adult*

Scaling

- `StandardScaler` (default), `RobustScaler` (outliers), `MinMaxScaler` (bounded). Trees don't need scaling.

7. Class Imbalance

To be expanded in Telco Churn & CTG projects.

Tool	Idea
<code>class_weight='balanced'</code>	Penalize minority errors more (built into many sklearn models)
<code>scale_pos_weight</code>	XGBoost equivalent = n_{neg} / n_{pos}
SMOTE	Synthesize minority samples (use <code>imblearn.Pipeline</code> ; fit on TRAIN folds only)
Threshold tuning	Lower the 0.5 cutoff to catch more positives
Metric switch	Use PR-AUC / macro-F1 , not accuracy (accuracy is misleading when imbalanced)

India Cancer note: SMOTE ≈ no-balancing on weighted-F1 there; `scale_pos_weight` hurt it. **Always test, don't assume.**

8. Model Selection

Always start with a baseline

- `DummyClassifier(strategy='most_frequent')` / `DummyRegressor(strategy='mean')`.
- If you don't clearly beat it, nothing else matters.

Roster by family (small/medium tabular)

Family	Models
Linear	LogisticRegression, Ridge, Lasso, ElasticNet
Distance	KNN
Probabilistic	GaussianNB
Margin	SVM (fine on small data; slow on large)
Single tree	DecisionTree
Bagging	RandomForest, ExtraTrees
Boosting	GradientBoosting, XGBoost , LightGBM , AdaBoost

- Compare with **StratifiedKFold** CV (classification) / **KFold** (regression).
- **Key lesson:** on small clean tabular data, **Logistic/Linear often beats boosting**. Complexity must earn its place.
- Exclude SVM/KNN on very large data (too slow).

9. Hyperparameter Tuning

Search	Use when
<code>GridSearchCV</code>	Small grid, fast model (exhaustive)
<code>RandomizedSearchCV</code>	Large grid / slow model (n_{iter} samples)

Common knobs

- **LogisticRegression**: `C` (logspace -3→2), `penalty` (l1/l2/elasticnet), `solver` (liblinear/saga), `class_weight`.
- **RandomForest**: `n_estimators`, `max_depth`, `min_samples_leaf`, `max_features`.
- **XGBoost**: `n_estimators`, `learning_rate`, `max_depth`, `subsample`, `colsample_bytree`.
- Param naming in pipeline: `clf__C`, `clf__max_depth`.

Honest note: on small clean data, tuning often moves the score by only ± 0.01 . Confirming the default is good IS a valid result.

■ **Selection tax**: `grid.best_score_` is optimistically biased (winner's curse). The honest fix is **nested CV**.

10. Evaluation & Metrics

Classification

Metric	Use
Accuracy	Only if balanced
Precision / Recall	Recall = catch positives (medical!); Precision = avoid false alarms
F1 / macro-F1	Imbalanced; macro-F1 for multiclass (rare classes count equally)
ROC-AUC	Ranking quality, threshold-free (good for model selection)
PR-AUC	Better than ROC when positives are rare
Confusion matrix	See <i>*which*</i> errors
Calibration curve	Do predicted probabilities mean what they say?

Regression

- **RMSE** (penalizes big errors), **MAE** (robust), **R²** (variance explained).
- Always beat DummyRegressor (mean) → $R^2 > 0$.

Decision threshold

- Default 0.5 is arbitrary. Choose deliberately:
- **Youden's J** = $\text{argmax}(\text{TPR} - \text{FPR})$ (balanced).
- **Cost-based** = pick threshold minimizing business/clinical cost (FN vs FP weighting).

11. Rigor & Honesty Checks

The "Chunk 7" template — run these at the end of every project.

- **Distribution, not a point** — `RepeatedStratifiedKFold(5×10)`; report **mean ± std**. A single hold-out (esp. small test set) is one noisy draw.
- **Error bars on the bake-off** — top models within 1 std = statistically tied → pick on interpretability/speed.
- **Right metric + chosen threshold** — report sensitivity/recall, not just accuracy; set the operating point on purpose.
- **Domain check** — what is the model **really** using? (P1: demographics-only AUC ~0.65 vs full ~0.90 → it's a diagnosis aid, not a screener.)
- **Learning curve** — converged gap = data-saturated (more data won't help); open gap = get more data.
- **Validation curve** — plot train vs val over a key param to see the overfit→underfit arc.
- **Distrust too-good scores** — on small/leaky data, >0.95 usually = a bug, not genius.

12. Interpretability

Tool	Model	Reading
Coefficients	Linear	+ pushes to class 1, - to class 0; magnitude = strength (standardize first)

Tool	Model	Reading
Feature importance	Trees	impurity-based (biased to high-cardinality)
Permutation importance	Any	shuffle a feature, measure score drop — multivariate, catches interactions
SHAP	Any (TreeExplainer for trees)	per-prediction contributions; use <code>shap.Explainer</code> (TreeExplainer breaks on new XGBoost)

Cross-check: do the top features match what EDA/stat tests flagged? If L1 penalty wins, weak features get coefficient = 0 (auto selection).

13. The Golden Rules

- **Split before you learn anything** (impute/scale/encode after split, inside Pipeline).
- **Beat the dumb baseline** or nothing else matters.
- **Right metric** — not just accuracy (macro-F1 / PR-AUC / RMSE).
- **Significance $\neq$ strength** — pair every p-value with an effect size.
- **Distrust scores that are too good** — near-perfect on small/leaky data = methodology bug.
- **Measure, don't assume** — test transforms/resampling/tuning with CV; keep only real gains.
- **Complexity must earn its place** — prefer the simpler model when scores tie.

14. ■ Techniques Log

Append every new trick here with the project + date it was learned.

Date	Project	Technique	One-line summary
2026-06	India Cancer	Leakage column detection	Dropped <code>Survival_Months</code> (consequence of target) → killed fake 97% accuracy
2026-06	India Cancer	Pipeline + ColumnTransformer	OHE + scale as one leakage-safe object
2026-06	India Cancer	SMOTE vs class_weight test	Resampling didn't help weighted-F1 here — always test
2026-06	P1 Heart	Grouped imputation (custom transformer)	Fill <code>ca/thal</code> by age-band median/mode, leakage-safe
2026-06	P1 Heart	Skew handling decision	log1p vs PowerTransformer; trees are skew-immune
2026-06	P1 Heart	Combined plot + stat test	Bar/box chart with chi-square/t-test annotation on same figure
2026-06	P1 Heart	Permutation importance	Multivariate feature value; beats single correlation bar
2026-06	P1 Heart	Rigor & Honesty chunk	Repeated CV distribution, error bars, threshold, learning/validation curves
2026-06	P1 Heart	Domain check (demographics-only)	Quantify how much skill comes from "downstream" features
2026-06	Telco Churn	Hidden missingness	Blank/whitespace strings aren't NaN — audit both; TotalCharges trap
2026-06	Telco Churn	Domain imputation	<code>tenure=0</code> → <code>TotalCharges=0</code> (meaning-based, not median)

Date	Project	Technique	One-line summary
2026-06	Telco Churn	class_weight vs SMOTE	class_weight='balanced' simplest; both mostly shift the boundary, PR-AUC ~flat
2026-06	Telco Churn	PR-AUC for imbalance	average_precision over ROC-AUC when positives are rare
2026-06	Telco Churn	Threshold tuning	F1-optimal & target-recall cutoffs; the real recall lever
2026-06	Telco Churn	Calibration	CalibratedClassifierCV fixes probabilities; class_weight distorts them (Brier ↓, ROC-AUC unchanged)
2026-06	Telco Churn	Leakage: row-by-row vs learns-from-data	fixed-bin/count/flag safe before split; target/mean/std/OHE must be inside pipeline
2026-06	Telco Churn	Decision-tree viz for interactions	shallow plot_tree shows the splits the model picks — beats manual interaction hunting
2026-06	Telco Churn	Systematic interaction scan	rank Contract×every feature by within-group churn spread
2026-06	Telco Churn	Tuning often negligible	gains come from metric/imbalance/threshold, not hyperparameters
	(next)		

15. ■ Scoreboard

Best honest result per project — for sanity-banding future work.

Project	Task	Metric	Score	Baseline	Notes
India Cancer	Binary (Alive/Deceased)	Weighted F1	~0.71	—	After dropping leakage; GB ≈ LR ≈ XGB
P1 Heart Disease	Binary (disease)	ROC-AUC	~0.84–0.90 (full)	Dummy ~0.54	Demographics-only ~0.65–0.70; models statistically tied
Telco Churn	Binary (imbalanced)	PR-AUC / ROC-AUC	~0.65 / ~0.84	Dummy recall=0	recall dialed to 0.79 via class_weight+threshold; tuning negligible
Ames Housing	Regression	RMSE / R ²	*TBD*		
Fetal Health (CTG)	Multiclass	macro-F1	*TBD*		
Medical Insurance	Regression	RMSE / R ²	*TBD*		
Covertypes	Multiclass (big)	macro-F1	*TBD*		
Energy Efficiency	Multi-output reg	RMSE / R ²	*TBD*		
Glioma Grading	Binary (LGG vs GBM)	ROC-AUC / F1	*TBD*		UCI #759; clinical + gene-mutation features; ref MDPI IJMS 2022

End of playbook. Keep it close, keep it growing.